IILM UNIVERSITY, GREATER NOIDA

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

PROJECT TESTING REPORT

Project Title: Predictive Maintenance System for Data Centre Servers

1. Project Details

Project ID	26P_3CSE276-Predictive Maintenance System for Data Centre Servers
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2. Purpose of Testing

This report records the checks run on the Predictive Maintenance System for Data Centre Servers. The system uses server monitoring data (CPU usage, memory usage, temperature and disk usage) to identify potential server issues and give a predictive maintenance indication or alert.

Note: All eight test cases were executed against the implemented code (pm.py and run.py) and the actual results below are the observed outputs. The dataset used is synthetic (generated by the project's script, seed 42), not real data-centre telemetry. All figures below were measured by running the code on that dataset; results on real server data will differ.

3. Testing Scope

- Input server-monitoring data is accepted and processed.
- Data validation for missing, invalid, or out-of-range values.
- The prediction component produces an output for valid input.
- Potential failure/maintenance conditions are identified according to the implemented model.
- Results are returned in the intended format (a status, prediction and probability; no graphical interface is part of this implementation).

4. Test Cases and Results

Test ID	Test Scenario	Expected Result	Actual Result	Status
TC-01	Provide a valid server data record (CPU, RAM, temperature, disk usage).	System accepts and processes the record.	Record validated and accepted as numeric values.	Pass

Test ID	Test Scenario	Expected Result	Actual Result	Status
TC-02	Provide a record with a missing required field.	System handles/reports missing data without crashing.	Rejected cleanly: Missing required field: temperature	Pass
TC-03	Provide a non-numeric value for a numeric parameter.	System validates the input and reports an appropriate error.	Rejected cleanly: Non-numeric value for cpu_usage: 'abc'	Pass
TC-04	Provide values within the dataset's valid range.	System processes the values correctly.	All 500 in-range records accepted and processed.	Pass
TC-05	Run prediction on a valid input record.	System returns a prediction or maintenance status.	Returned NORMAL (p=0.0416).	Pass
TC-06	Provide a record representing abnormal/high server readings.	System identifies the condition according to its implemented model/rules.	Returned MAINTENANCE ALERT (p=0.9926).	Pass
TC-07	Check the displayed prediction/result.	Output is displayed in the intended format and matches model output.	Output has prediction, status and probability; matches direct model output.	Pass
TC-08	Run the system with multiple records.	System processes records without unintended failure.	Processed 1000 records in 1.62s; 998 predicted, 2 invalid records reported without stopping the batch.	Pass

Summary: 8 of 8 test cases passed.

5. Test Environment

Hardware	Cloud virtual machine, 2 vCPU, 7 GB RAM
Operating System	Linux 6.18.44-fc-v185 (cloud VM)
Programming Language / Tools	Python 3.13.18; scikit-learn 1.9.1, pandas 3.0.5, NumPy 2.5.3
Dataset	server_telemetry_synthetic.csv (synthetic, 5,000 records, generated by pm.py with seed 42)

6. Summary and Conclusion

All 8 test cases passed. Invalid inputs (missing, non-numeric and out-of-range values) were rejected with clear error messages without crashing, valid records were processed, an abnormal record (CPU 98%,

memory 95%, temperature 95 °C, disk 95%) was flagged as a maintenance alert, and a batch of 1,000 records containing 2 invalid entries completed without stopping. No defects were found in this run.

Limitations: testing used synthetic data and a code-level test harness; it did not cover real server telemetry, a graphical interface, or performance under live load.